Mortality Patterns: Retrospective Estimation and scenario-based forecasts- **GitHub Documentation**

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Introduction & Overall Description

The GitHub repository 'mortality_estimation' has been created to examine mortality rates in different countries. The main objectives of this study are to:

- Project or **estimate the crude death** rate using measured values of various predictor variables,
- **forecast scenarios** for the next five months based on the predictions of these predictor variables.

To achieve these two main objectives, the GitHub repository is organised into **10** separate folders, as shown in Figure 1. The following section will first guide you through the libraries and versions of R required for successful execution of the different steps. After that, we'll dive into a detailed exploration of the different folders, most of which are made up of four main parts: data, results, visualisation and code.

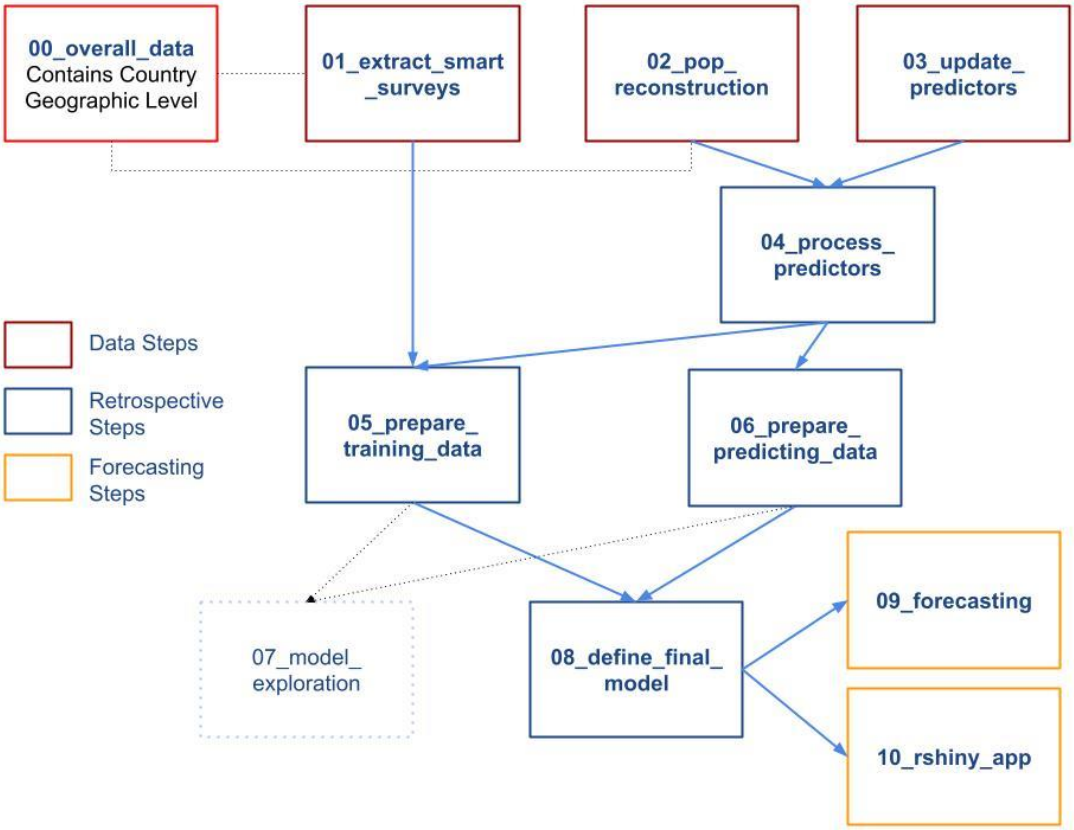

Figure 1. Visualization of the Github Structure

Note: If you only wish to view and experiment with the **RShiny app** for forecasting scenarios, please refer to the next section (Packages, libraries and versions) to ensure that you have the necessary packages to launch the RShiny app. Then go to the folder entitled "10_rshiny_app" and follow the instructions provided in the following video:

https://lshtm.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=7ea184cf-2028-4554-867a-b0d901059162

Packages, Libraries & Versions

To successfully execute

Data Steps

Retrospective Steps

Forecasting & RShiny Steps